

ARUN KUMAR SINGH

San Diego, CA, USA

✉ arunsingh.in@gmail.com  [linkedin.com/in/arunsingh23/](https://www.linkedin.com/in/arunsingh23/)  github.com/arunsingh  medium.com/@aruns89-techblog  dyota.substack.com-newsletter

Professional Summary

Business focused, Customer Obsessed engineering aficionado with cross-functional 16+ years of Industry, Start Ups R&D experience in ICT, AIops, FinOps, healthcare, Industrial IoT energy & utilities, specialised software product tech development, SRE, Cloud Infra, architecture, Security, Production support, Operations, Incident Management.

Education

Harvard University, J.F Kennedy School of Government

Public Leadership Credential

Feb 2026 – Ongoing

Cambridge, MA, USA

Krishna Institute of Engineering & Technology

Bachelor of Technology in Information Technology

Aug. 2005 – Jun 2009

Delhi NCR, India

Technical Skills

Languages: Python, Java, C, Qiskit, PennyLane, Javascript, Typescript, *nix shell scripting, Golang, Rust, Ruby, SQL

Developer Tools: VS Code, Eclipse, Google Cloud Platform, PyCharm, AWS Cloud9

Technologies/Frameworks/Toolkits/DB/API: Ubuntu MaaS, OpenStack, Nautobot, Datadog, ExtraHop, Puppet, Ansible, Docker, Kubernetes, Packer, Terraform, Spinnaker, Jenkins, New Relic, Redis, AWS, GCP, Azure, ELK, LitmusChaos, DNS, SSL/TLS, TCP/IP, HTTP(S), $\text{\LaTeX}$, Gatling, GIT, Kafka, Grafana, perf, flamegraph, kvm, JMeter, K6, Flask, Pager-duty, Django, MySQL, MongoDB, PostgreSQL, Clickhouse, Bezel, Slurm, ArgoCD, Mesos, LangChain, N8n, Llama, GPT, AutoGen, AGI, Cirq

Hardware Platform(s): NISQuantum, RaspBerry Pi, Arduino, Bare-Metal, Beagle, Custom PCBs.

Policies : HIPAA, CIS, SOC2, PCI-DSS, GDPR, ITIL, DPDP.

Professional Certifications

Languages, Domains: NVIDIA AI Infrastructure and Operations(2025), Google Cyber Security (2024), Python (2018-UMichigan), Unix shell scripting (2019), AWS (2019, 2021, 2023), GCP (2019), Chaos Engineering (2024)

Platforms: LINUX (2019, 2023), K8s (2023), Docker (2023), Linkerd (2023), Istio (2024), Prometheus (2023)

Frameworks & Leadership: Harvard University - Exercising Leadership: Foundational Principles (2024), ITIL Service Management (2024), Google Cloud - SRE: Measuring & Managing Reliability (2019), Sumologic - Log, Search, Analytics (2021)

Professional Experience

Godspeed Pte Ltd

VP & Sr.Principal Engg - SRE, Production AI Infrastructure & Platform Engineering

Nov 2022 – Present

San Diego, CA, USA

- People Management: IC role:: 30:70, Platform owner for multiple projects. **Team size: 22**, tech stack: AWS, GCP, Azure, fastly, Prometheus, Grafana, OTEL, Python, Golang, Typescript, JS, Docker, Kubernetes, Ansible, GitLab, ArgoCD, Kafka, Debezium, Airflow, Terraform, helm, Airbyte, Linkerd, Spark, Snyk, Blackduck, Gatling, Extrahop.
- **Key Accomplishments**
- **Architected GPU cluster topology and AI inference infrastructure** for multi-DC deployments, optimizing compute-node placement, workload scheduling, and GPU density for large-scale **LLM training and inference pipelines**.
- Shipped and led architecture of a **Nautobot Agentic AI to maintain asset lifecycle management for both On-Prem and Cloud Assets**, Gen AI **LLM-based platform** integrating multiple cloud vendors, dev tooling, coding instrumentation, observability, ProdOps, **enhancing developer productivity by 30X using IDP**.
- Built **Smart Inferencing system intelligence engine to process token AI workloads** consisting of Smart Queues, Dynamic Model Router, Custom Model Parameters, Tuned Nodes that turn compute into intelligence and outcomes.
- Shipped Quantum Computing Algorithms, Quantum-kernel-OS, Quantum Simulation platform, Developed PoC, a Linux-based traffic generator for **network performance testing**, Shift left **QA automation using IaC for cloud-based workloads**.
- Fundraising, sales, customer onboarding PoCs, **team building**, training bootcamps. GTM strategies resulting in 5X customer MoM. CASB, PCIDSS, HIPAA, NIST CSF, GDPR GRC **Security Audits (AppSec, InfoSec, Network Security)**.

Stealth Startup - Inductive Charging EV Mobility

Founder - SRE, Product and Platform Architecture

Nov 2021 – Oct 2022

Canada, India, Taiwan, Singapore

- People Management: IC role:: 10:90, Backend, platform, hardware owner for multiple projects. **Team size: 2-10**
- **Key Accomplishments**
- Bootstrapped a stealth startup focused on EV mobility, developing an inductive charging prototype for product-market fit evaluation. Despite progress, commercialization with OEMs was hindered by hardware optimization challenges.
- Created inductive charging hardware prototype through contract manufacturing in SE Asia.
- Led software platform for EV mobility, Battery Observability, health, OEM integration, Shift left Approach **SRE, performance benchmarking, tooling based automation across Code, Build, Test, Release, Deploy, Monitoring, Observability.**

NextGen Healthcare Inc (formerly QSI Inc)

Aug 2019 – Oct 2021

Sr. Staff Site Reliability Engineer

San Diego, CA, United States

- People Management: IC role:: 30:70, Backend, platform, infrastructure owner for multiple projects. **Team size: 5-68**, tech stack: AWS, fastly, Datadog, Sumologic, Python, Java, Docker, Kubernetes, Puppet, GitLab, opa, atlantis, Jenkins, Kafka, Databricks, Kong, Terraform, Airbrake, Linkerd, Oracle 11, MySQL, Redis, Airbyte, VMWare VSphere.
- **Key Accomplishments**
- Enhanced developer workflow and productivity within a **cross-functional team** by creating tools to automate and manage multi-cloud infrastructure, cutting operational overhead by a third. On-premises DC workload to Cloud migration.
- Developed a **performance benchmark suite testing tool** focused on improving 5x Reliability. Shipped product-wide Instrumentation, enterprise monitoring and reporting tools **for apps used by 4 million users.**
- Built PaaS abstractions over infrastructure for reducing multi-cloud deployment from a day to under an hour. Re-architected NGO BI Reports Infra to **unlock USD 50M customer subscriptions.**

MachinePulse Tech Pvt Ltd (now MahindraTeqo)

Oct 2017 – Feb 2019

Principal Architect - SRE, Product & Platform Engineering

Bangalore, India

- People Management: IC role:: 40:60, Backend, platform, hardware owner for multiple projects. **Team size: 8-45**, tech stack: AWS, on-prem, fastly, Sensus, ELK, Python, Java, Docker, Kubernetes, Puppet, GitLab, Jenkins, Kafka, Spark, Terraform, Sentry.
- **Key Accomplishments**
- Launched leading on-premise monitoring solution for Solar industry within Industrial IoT domain.
- Scaled system 8x to 3 million writes/sec across 700+ installations via event-driven architecture migration & **PoC, Implemented litmus Chaos engg tool.**
- Designed and implemented a tool for **load and performance testing of HTTP API** backend services.
- **Built support org and led a 30-member cross-functional team** of PM, devs, QA, and DevOps.

Infinite Forest, Inc

Oct 2016 – Aug 2017

Sr. SRE, Platform Engineering

Bangalore, India

- Shipped the back-end engine for machine learning-based matrimonial product. Created robust, fault-tolerant platform, built cross-platform 24X7 centralized monitoring.

Livestream Technologies Pvt Ltd (now Vimeo, Inc)

Jun 2016 – Sept 2016

Sr. Site Reliability Engineer, Platform & Product Group

Bangalore, India

- Managed scalability and uptime for a live video streaming platform on 4550 servers through automated system design, monitoring, and operational tools development.

Shadowfax Technologies Pvt Ltd

Oct 2015 – Jan 2016

Tech Architect - Backend Infrastructure & DevOps

Bangalore, Karnataka, India

- Created On demand hyperlocal delivery product using Python, Django framework.

Indian Institute Of Technology Delhi

Apr 2010 – June 2015

Sr. Computer Scientist

New Delhi, India

- Worked with IIT Delhi & incubation based start up **Gramvaani Ltd**, to understand **the State of Internet Connectivity in Rural Areas**. Performing Network auditing & performance measurements for wireless service providers. Built technology Infrastructure and deployed technology suites to rural field locations under e-governance initiatives of the Government of India.
- Multitouch Surface Table: Implemented TUIO protocol & developed a graphical Zoomable User Interfaces (ZUIs) by hacking Python(pyMT) and Java Library(MT4j) APIs, hacked an TFT screen to create 3D interactive table with Acrylics mask and also designed 100 inches interactive wall with fiducial marking that enables device to detect object.

Sun Microsystems, Inc

June 2009 – Mar 2010

Member Technical Staff

India

- Worked on Sun Cluster infrastructure on solaris kernel, troubleshooting solaris kernel crash dump analysis and subsystems feature viz. VM, ProcFs and boot up.

Sapro Robotics India Private Limited

Feb 2008 – Jan 2010

Co-founder & System Engineering

Ghaziabad, U.P. India

- Invented Surgical robots for healthcare, UAVs, and mobile robots for the defence sector.

Open Source and Side Projects

Apache Airflow, Cilium & Linkerd

2022 – Present

OSS

CNCF

- OSS Contribution since 2022 to Apache Airflow, CNCF Cilium, Linkerd **Triage issues, PRs.**

Open Source Software Contribution, Leadership, Speaking

Google GDG Cloud Bangalore MeetUp

Mar 2018 – Present

Community Founder & Tech Evangelist

Google

- **Contributor & Maintainer** for **Stanford University** based Eyepatch project and Creator for Elongpower.
- Managed executive board of 4 members and ran weekly meetings to oversee progress in essential parts of the chapter.
- Led chapter of 10000+ members to work towards goals that improve and promote tech community tech talks.

REFERENCES

Available on Request.